Assignment 3.

0. Exercise. Variance-covariance method for VaR & ES in linear portfolio: *simple example of data mining*
At the end of the 24th of July 2012 compute daily VaR and ES for an equally weighted equity portfolio with Inditex, BASF, LVMH, with

- α=95%;
- a 2y estimation using the dataset provided;
- Notional €10 Mln;
- via a Gaussian parametric approach.

[Hint: Pay attention that the trading days of the different stocks are not the same. Add previous day value in case of missing share price.]

1. Case study: Historical (HS & WHS) Simulation, Bootstrap and PCA for VaR & ES in a linear portfolio
At the end of the 24th of July 2012 an asset manager due to different rules on three company units (portfolios) has to compute risk measures according to the following rules (α=99%):

a. Portfolio 1 with: Eni (18K shares), Telefonica (25K Shares), Eon (15K Shares), Daimler (9K Shares). Compute daily VaR and ES with a 2y estimation using the dataset provided via a Historical Simulation approach and a Bootstrap method with 200 simulations.
b. Portfolio 2 with equally weighted equity: Vivendi, AXA, Enel, Volkswagen and Schneider. Compute daily VaR and ES with a 2y estimation using the dataset provided via a Weighted Historical Simulation approach with lambda = 0.98.
c. Portfolio 3. An equally weighted equity portfolio with the first 25 shares (in the Matlab function "underlyingCode.m" provided) in the EUROSTOXX 50 (up to France Telecom included). Compute 10 days VaR and ES with a 2y estimation using the dataset provided via a Gaussian parametric PCA approach. What is the minimum value of n principal components in the Gaussian parametric PCA approach to have an error less than 5% with respect to full Gaussian approach for 10 days VaR with a 2y estimation using the dataset? And less than 1%? Notional €15 Mln. Comment the results.

For the three portfolios he checks results' order of magnitude via a Plausibility check.

2. Exercise: Full Monte-Carlo and Delta normal VaR
At the end of the 15th of February 2008 consider a portfolio formed by stocks of Generali for 1,164,000 Euro and the same number of put options with expiry on the 18th of Apr 2008, with strike 28.5 Euro and volatility equal to 22.3% (dividend yield of 5.1%). Volatility and dividend yield are yearly quantities. Compute a 1dd/99% VaR via a Full Monte-Carlo and a Delta normal approaches (only delta term). Use a 2y Historical Simulation for the underlying.
Can you improve the Delta normal VaR? How?
Why the Full Monte-Carlo can be numerical intensive for an exotic derivative that cannot be priced via a closed formula?

3. Case Study: **Pricing in presence of counterparty risk**
On the 15th of February 2008 at 10:45 C.E.T. bank XX buys form ISP a 5y Cliquet option for a 45 MIO € notional. Option yearly payoff (annual bond) at payment date is

$$[S(t_i) - S(t_{i-1})]^+.$$

ROBERTO BAVIERA
POLITECNICO MILANO
E.Mail: financial.engineering.polimi@gmail.com

The option is on an equity stock (with no dividends) and constant volatility 19%. What should be the correct price? At what price ISP would try to sell it?
[Hint: consider the dynamics of the underlying not of the corresponding forward]

4. **Case Study**: MBS Pricing
On the 15th of February 2008, the reference portfolio of the SPV *Cayman II* has a total notional of € 1 bn: this portfolio can be considered homogeneous. Assume for simplicity that mortgages provide a single payment at the end of the interest period T equal to three years and defaults are independent from interest rates; for the period T the default probability of each mortgage is p=5%, correlation is 40% and an average recovery of 20% for each mortgage.

a. For the mezzanine Tranche (with subordinations -detachment points- K_d 5% and K_u 9%), estimate the impact of the hypothesis of Large Homogeneous Portfolio for I = 400 for the Vasicek model. Show that the price of the Tranche in relative terms (as a percentage of tranche face value) is well described by the Kullback-Leibler (KL) approximation in the range I = (10, 2 10^4), where the LHP approximation holds. It is required to plot the price varying I (in log scale in the abscissa), with

i) the exact solution (up to an I that your computer allows to obtain a price);
ii) the approximate solution;
iii) the LHP solution.

[Hint: Use a grid for I similar to the one in the document and plot prices in % in log-lin scale]

b. [facultative] Price the Equity tranche with detachment points K_u 5% varying I in the same range considered at the previous point. Is the KL approximation adequate? How can you modify it?

Name of some Matlab functions:

nchoosek(I,n):	Newton Binomial coefficient
norminv(alpha):	Inverse normal CDF
normpdf(x):	pdf normal
quadgk:	for numerical integration

Function signatures
[ES, VaR] = AnalyticNormalMeasures(alpha, weights, portfolioValue, riskMeasureTimeIntervalInDays, returns)
[ES, VaR] = HSMeasures(alpha, weights, portfolioValue, riskMeasureTimeIntervalInDays, returns)
[ES, VaR] = WHSMeasures(alpha, lambda, weights, portfolioValue, riskMeasureTimeIntervalInDays, returns)
[ES, VaR] = PCAMeasures(alpha, numberOfPrincipalComponents, weights, portfolioValue, riskMeasureTimeIntervalInDays, returns)
VaR = PlausibilityCheckVaR(alpha, weights, portfolioValue, riskMeasureTimeIntervalInDays, returns)
VaR = FullMonteCarloVaR(alpha, numberOfShares, numberOfPuts, stockPrice, strike, rate, dividendYield, volatility, TTMinYears, riskMeasureTimeIntervalInDays, returns)
VaR = DeltaNormalVaR(alpha, numberOfShares, numberOfPuts, stockPrice, strike, rate, dividendYield, volatility, TTMinYears, riskMeasureTimeIntervalInDays, returns)

Pls remember that

- all vectors are column vectors.
- returns are daily returns;
- In FullMonteCarloVaR & DeltaNormalVaR: rate, dividendYield, volatility, TTMinYears are yearly.

ROBERTO BAVIERA
POLITECNICO MILANO
E.Mail: financial.engineering.polimi@gmail.com